Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE



Digital Forensics & Biometrics

A.A. 2026/2027

Lecture 08 – Multimedia evidences

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INTRO ON MULTIMEDIA FORENSICS

Intro video from IEEE Signal Processing Society

THE PROCESSING HISTORY OF A MULTIMEDIA CONTENT



Real scene



Acquiring device



Storage

Post-acquisition editing

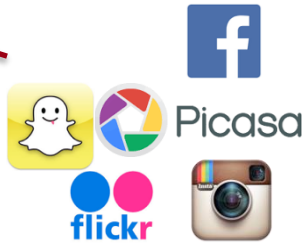






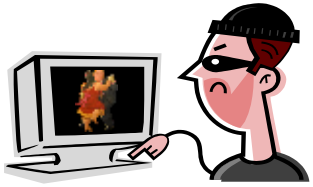
User 1

Download



Upload





New upload





User 2

PROBLEMS:

For a given image/video/audio file:

- Who took it? Who created it?
- What is its origin?
- Was it modified? If so, which changes were operated? How much of the original content was altered?
- Can I trust its content?

Aside to these, additional questions can be posed:

- What this multimedia content tells me about the scene where it was acquired?
- Is it possible to relate it to other contents?
- Was the software/device who generated/took it legal?

AUTHENTICATION OF DATA

Given a multimedia evidence, was it acquired by that device ?
At that time?



?



- Validation of evidences
- Tracking of the origin of illegal data
  - Child pornography
  - Material promoting illegal activities (terrorism, …)

Who owns the data?

- Watermarking

Was it manipulated? If so, in what parts? How much of the original content has been changed?

DATA ENHANCEMENT

Tools for image/video enhancement in multimedia forensics

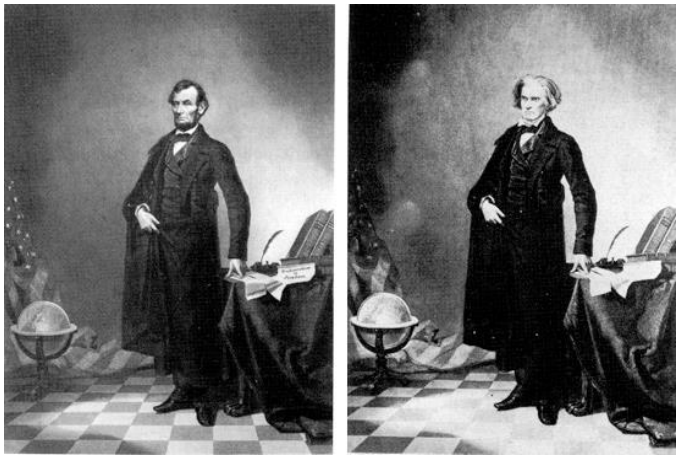
Images/videos/audios are usually taken in adversarial conditions (e.g., video surveillance at night, eavesdropping in a noisy environment)

- Need to process evidences so that quality is enhanced.
- Perform this in a forensically sound way
- Localize and isolate events/details from data
- Identify object/persons/time/locations



FAKE DETECTION

1860



Fake photos/videos/audios have been generated since the beginning of photography/music

- Artistic reason
- Frauds
- Manipulate public opinion
- Promote trends, aesthetic models

Need to detect manipulated data from real ones!





2007



2011



Last appeared in 2012

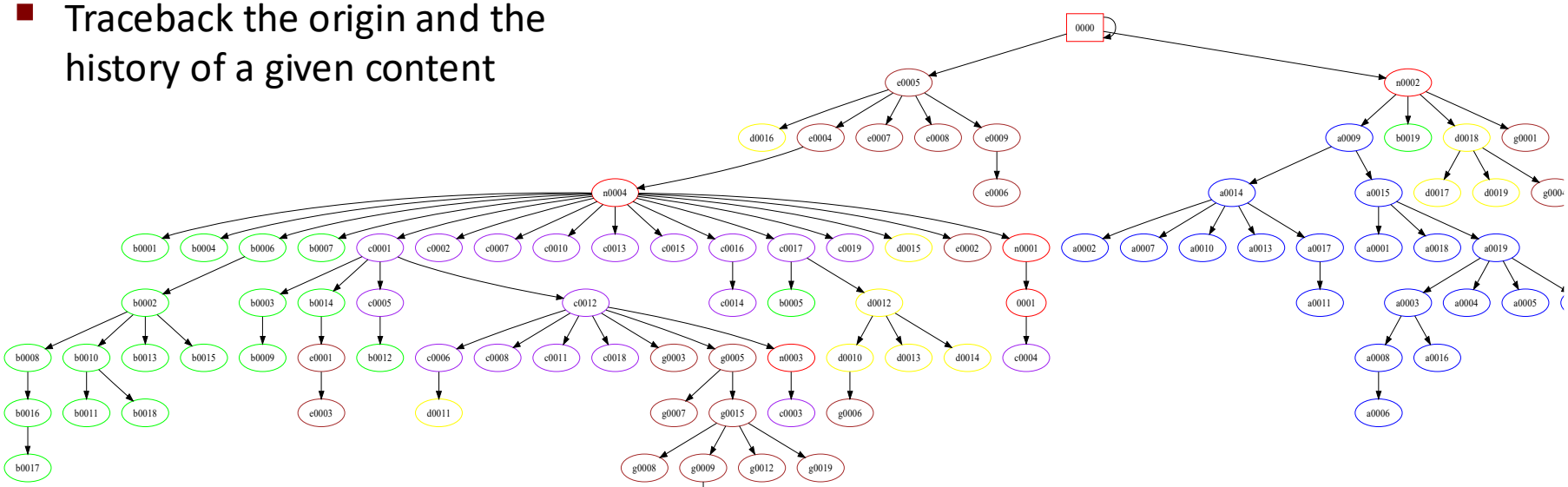


2014



TRACE THE ORIGIN OF A GIVEN CONTENT

- Often contents goes viral
- Multiple copies exists
- Traceback the origin and the history of a given content




ID h - Crop*


ID i - Other*


ID 0000 - Original


ID a - Balotelli*


ID b - Text Overlay*


ID c - Watermark*

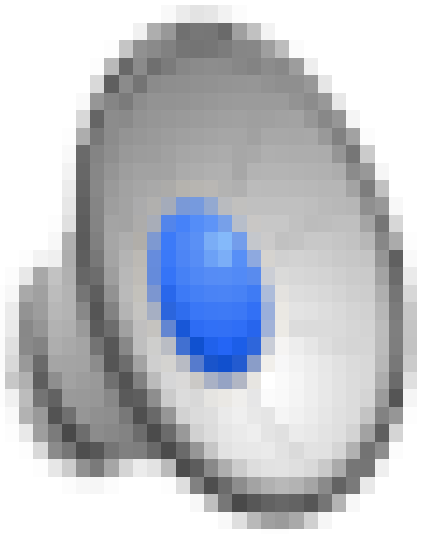

ID d - Face Swap*


ID e - Joystick*


ID f - Splicing*


ID g - Hat*

RECONSTRUCTING THE ORIGINAL SOURCE



OBSERVATIONS

- Lifetime of digital objects is unbounded
- Objects go through multiple processing steps
- Processing steps leave characteristic footprints

- Typically, footprints are deemed as "misprints"
- MF approach: footprint as an asset

Trace back the history of multimedia objects

HOW MULTIMEDIA FORENSICS TRIES TO ANSWER …

Multimedia contents can be analyzed under different aspects.

Most of the multimedia files are endowed with meta-data related to acquisition device, editing software, etc…

e.g. Exchangeable Image File Format (EXIF) tags

- Focal length/ exposure
- GPS location
- Sampling
- Editing software (e.g. Photoshop)



Forensic analyst can use them but ….



They can be easily removed …

Look for traces on the pixels of image/videos (or on samples for audio track)



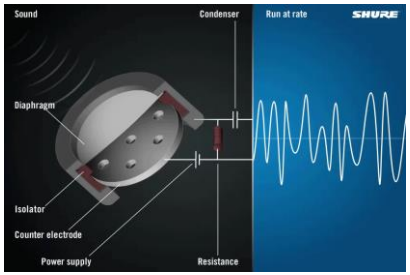
EXAMPLE

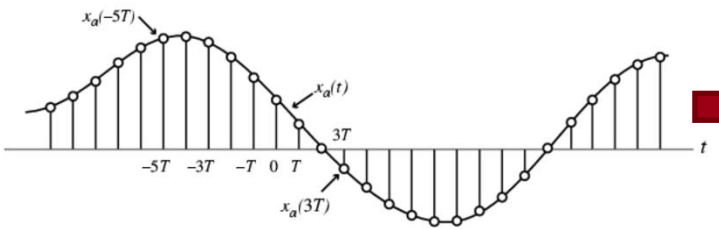
	Fb Web (alb hq)	Fb Web (alb lq)	Fb Web (single)	Fb And (album)	Fb And (single)	Tw Web	Tw And	G+ Web (normal saving)	G+ Web (zip)	G+ And (album)	G+ And (single)	Insta Web	Insta And (hq)	Insta And (lq)	Pin Web	Pin And	Snap And	WA Web	WA And	We Chat Web	We Chat And	Win Import	Flickr	Tele Web	Tele And	
Fb Web (alb hq)		55	50	85	90	90	90	100	95	100	100	90	95	95	85	100	100	100	80	95	90	100	95	90	80	Fb Web (alb hq)
Fb Web (alb lq)	55		60	90	90	90	90	100	100	100	100	90	95	95	85	100	100	100	95	95	90	100	95	95	80	Fb Web (alb lq)
Fb Web (single)	50	60		85	90	90	90	100	95	100	100	90	95	95	85	100	100	100	80	95	90	100	95	90	80	Fb Web (single)
Fb And (album)	85	90	85		80	85	85	95	90	100	100	85	95	95	100	95	100	95	90	90	85	95	90	95	90	Fb And (album)
Fb And (single)	90	90	90	80		90	90	100	95	100	100	90	95	95	95	100	100	100	80	95	90	100	95	95	90	Fb And (single)
Tw Web	90	90	90	85	90		50	90	90	90	90	90	85	85	90	90	100	90	90	85	85	90	85	90	80	Tw Web
Tw And	90	90	90	85	90	50		90	90	90	90	90	85	85	90	90	100	90	90	85	85	90	85	90	80	Tw And
G+ Web (normal)	100	100	100	95	100	90	90		100	55	85	90	85	85	100	100	100	100	100	100	100	100	100	100	100	G+ Web (normal)
G+ Web (zip)	95	100	95	90	95	90	90	100		100	100	95	95	95	100	100	100	100	80	80	90	85	80	100	90	G+ Web (zip)
G+ And (album)	100	100	100	100	100	90	90	55	100		80	90	85	85	100	100	100	100	100	100	100	100	100	100	100	G+ And (album)
G+ And (single)	100	100	100	100	100	90	90	85	100	80		95	95	95	100	100	100	100	100	100	100	100	100	100	100	G+ And (single)
Insta Web	90	90	90	85	90	90	90	90	95	90	95		50	80	85	90	100	100	95	95	90	100	95	95	80	Insta Web
Insta And (hq)	95	95	95	95	95	85	85	85	95	85	95	50		80	55	75	100	95	95	95	95	95	95	95	90	Insta And (hq)
Insta And (lq)	95	95	95	95	95	85	85	85	95	85	95	80	80		70	75	100	95	95	95	95	95	95	95	85	Insta And (lq)
Pin Web	85	85	85	100	95	90	90	100	100	100	100	85	55	70		60	100	100	95	100	75	100	100	100	80	Pin Web
Pin And	100	100	100	95	100	90	90	100	100	100	100	90	75	75	60		100	100	100	100	90	100	100	100	85	Pin And
Snap And	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100		100	100	100	100	100	100	100	100	Snap And
WA Web	100	100	100	95	100	90	90	100	100	100	100	100	95	95	100	100	100		100	100	100	100	100	100	100	WA Web
WA And	80	95	80	90	80	90	90	100	80	100	100	95	95	95	95	100	100	100		80	70	85	80	100	80	WA And
WeChat Web	95	95	95	90	95	85	85	100	80	100	100	95	95	95	100	100	100	100	80		85	65	50	100	90	WeChat Web
WeChat And	90	90	90	85	90	85	85	90	85	100	100	90	95	95	75	90	100	100	70	85		90	85	70	60	WeChat And
Win Import	100	100	100	95	100	90	90	100	85	100	100	100	95	95	100	100	100	100	85	65	90		65	100	100	Win Import
Flickr	95	95	95	90	95	85	85	100	80	100	100	95	95	95	100	100	100	100	80	50	85	65		100	90	Flickr
Tele Web	90	95	90	95	95	90	90	100	100	100	100	95	95	85	100	100	100	100	100	100	70	100	100		100	Tele Web
Tele And	80	80	80	90	90	80	80	100	90	100	100	80	90	85	80	85	100	100	80	90	60	100	90	100		Tele And

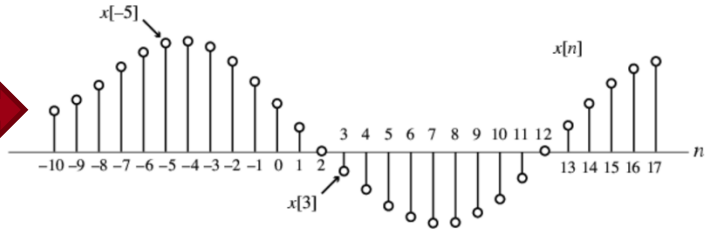
BASICS OF SIGNAL PROCESSING

After acquisition, digital signals can be modeled from time series or matrices of integer values

Audio example

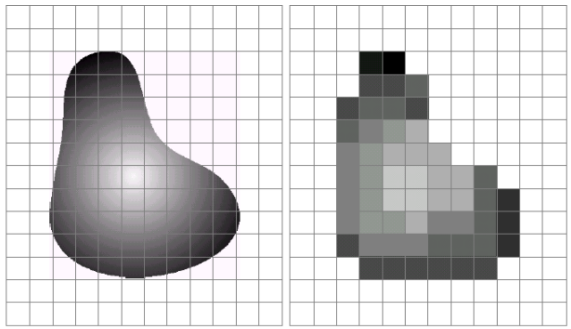


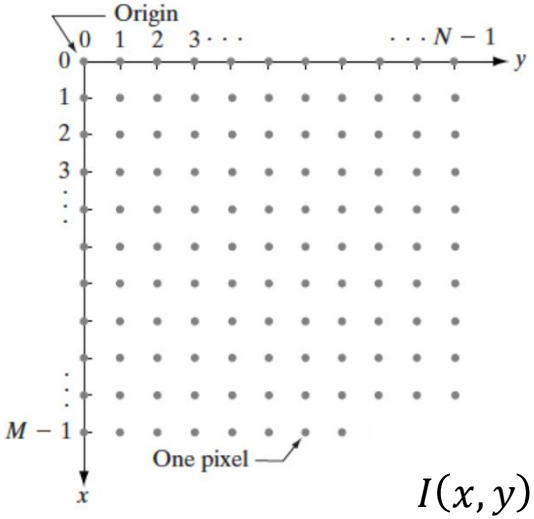




$$F_T = \frac{1}{T}$$ Sampling frequency $$x[n] = x_a(t)_{t\ =\ nT} = x(nT)$$

Image example



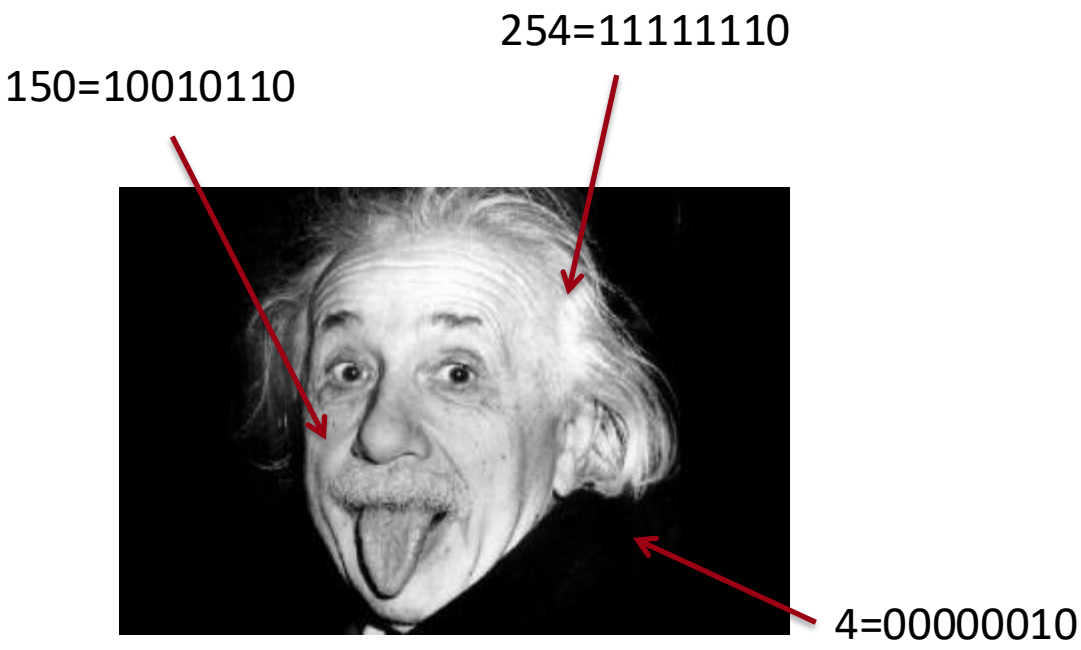


$$I(x,y) = \begin{bmatrix} I(0,0) & \cdots & I(0,N-1) \\ \vdots & \ddots & \vdots \\ I(M-1,0) & \cdots & I(M-1,N-1) \end{bmatrix}$$

QUANTIZATION

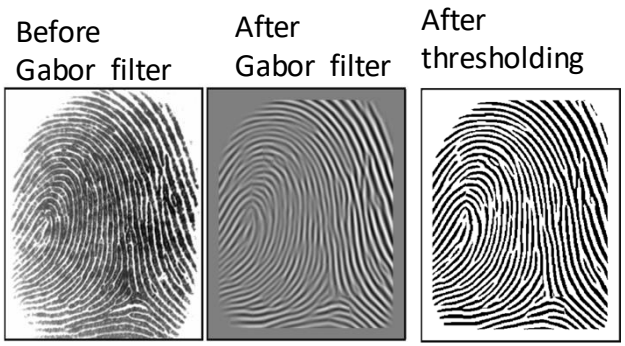
- Each sample is quantized and represented with a finite number of bits (8, 12, or 16 bits)
- Most images use 8 bits/sample: each sample is an integer from 0 to 255 depending on the intensity of the light; For color images, digitization is performed on the three components R,G,B
- Binary signals are present as well

Es.



BINARY SIGNALS

- Sometimes whenever shape and geometric details are involved, binary images are used
- Usually they can be obtained thresholding a standard image or applying some segmentation
- Examples: fingerprints, silhouettes for gait analysis





- The signal is a matrix whose cell can assume binary values {0,1}

$$I = \begin{bmatrix} I(0,0) & \cdots & I(0,N-1) \\ \vdots & \ddots & \vdots \\ I(M-1,0) & \cdots & I(M-1,N-1) \end{bmatrix}$$

$$I(x,y) \in \{0,1\}$$

EXAMPLES OF BINARIZATION (1/2)

☐ Thresholding

$$B(u,v) = \begin{cases} 1 & I(u,v) \geq T \\ 0 & I(u,v) < T \end{cases}$$

where T can be computed in different ways.

Otsu's method: minimize intra-class variance $\sigma_{Intra} = p_0\sigma_0 + p_1\sigma_1$ where p_0 (p_1) is the probability (percentage) that the pixel is lower (higher) than T. σ_0 (σ_1) is the variance for the class of 1s (0s).

$$\min \sigma_{intra} = \max \sigma_{inter} = \max(\sigma - \sigma_{intra}) = \max p_0 p_1 (\mu_1 - \mu_0)^2$$

1. Compute histogram and probabilities of $I(u,v)$
2. Set p_0, p_1 and μ_0, μ_1
3. **For** $T = 1, \dots, 255$
   compute the new p_0 (p_1) and μ_0 (μ_1)
   compute σ_{inter}
   If σ_{inter} is max, **then**
   $T_{opt} = T$



EXAMPLES OF BINARIZATION (2/2)

☐ Segmentation of foreground objects

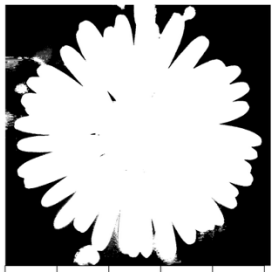
Simple case: define background and Foreground with respect to colors

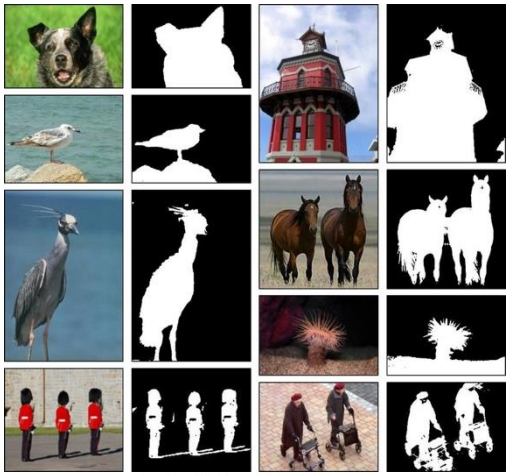
Background

Foreground



With knn refinement





Moe complex solutions: gradient based, watershed, using neural networks

OTHER BIT DEPTHS

- Genomic signal processing for DNA/RNA sequences

  Signal can be seen a function

$$\boldsymbol{s}[n]: \mathbb{R} \mapsto \{A, G, C, T\}$$



  or a set of 4 binary functions (Voss representation)

$$\hat{s}_1[n] = \begin{cases} 1 & \text{if } s[n] = A \\ 0 & \text{otherwise} \end{cases}$$

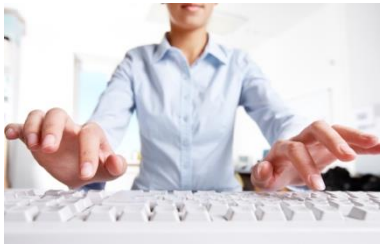
$$\hat{s}_2[n] = \begin{cases} 1 & \text{if } s[n] = G \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{s}_3[n] = \begin{cases} 1 & \text{if } s[n] = C \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{s}_4[n] = \begin{cases} 1 & \text{if } s[n] = T \\ 0 & \text{otherwise} \end{cases}$$

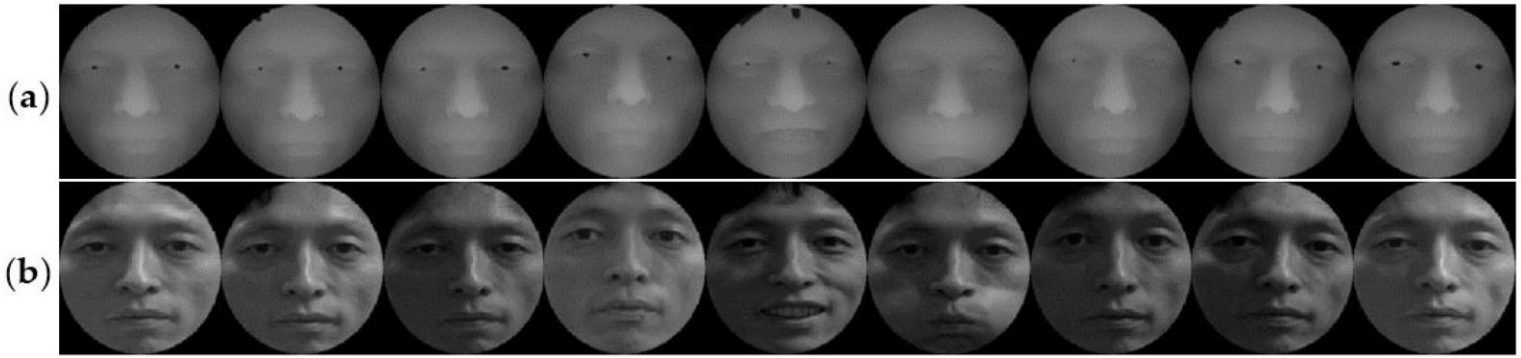
- Typing biometrics: user's typing patterns (speed, the duration of a single keypress, and how long it takes between releasing a key to pressing the next).

  signals taking values in the keyboard keys



DEPTH AND 3D SIGNALS (1/2)

- 3D sensors are employed as well
- The acquired signal is usually a depth map $D = [d(x, y)]$



- Depth value is an integer parameterizing the distance of the pixel from the camera itself.
- From depth representation it is possible to infer a 3D modelling of faces or body



DEPTH AND 3D SIGNALS (2/2)

A mesh model need to be specified by the following elements

- Vertex
- Edges
- Faces
- Normals

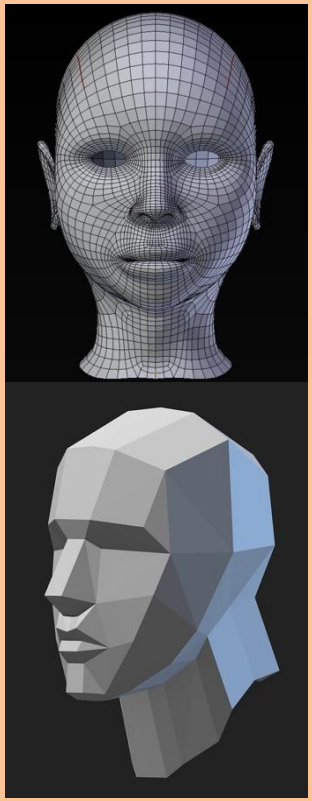
Types of meshes:

- Generic (rare)
- Triangular
- Rectangular

OpenGL use generic meshes,
But convert them into triangular

PLY format

```
ply
format ascii 1.0
element vertex 8
property float x
property float y
property float z
property uchar red
property uchar green
property uchar blue
element face 7
property list uchar int vertex_index
element edge 5
property int vertex1
property int vertex2
property uchar red
property uchar green
property uchar blue
end_header
0 0 0 255 0 0
0 0 1 255 0 0
0 1 1 255 0 0
0 1 0 255 0 0
1 0 0 0 0 255
1 0 1 0 0 255
1 1 1 0 0 255
1 1 0 0 0 255
3 0 1 2
3 0 2 3
4 7 6 5 4
4 0 4 5 1
4 1 5 6 2
4 2 6 7 3
4 3 7 4 0
0 1 255 255 255
1 2 255 255 255
2 3 255 255 255
3 0 255 255 255
2 0 0 0 0
```



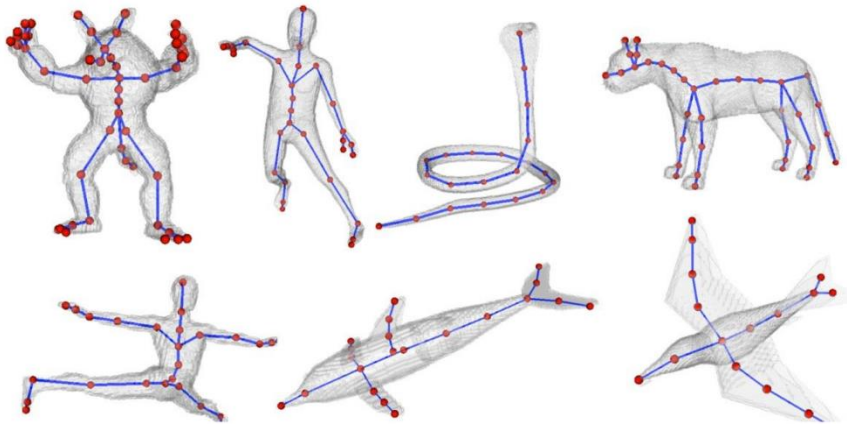
3D TIME SERIES

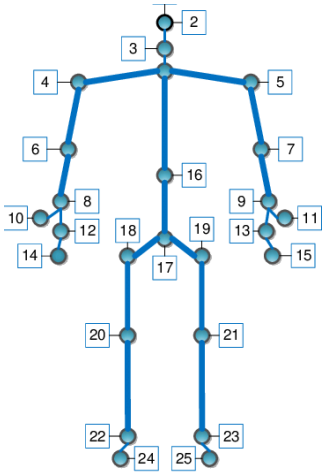
Whenever the identification involves the time modelling of a set of 2D/3D coordinates, n-dimensional time series can be involved

$$\boldsymbol{s}[n]: \mathbb{R} \mapsto \mathbb{R}^n$$

Example: **joints position in skelethon for gait modelling**

- A skeleton is a framework of rigid body"bones" connected by articulated joints
- Used as an (invisible?) armature for the articulated bodies





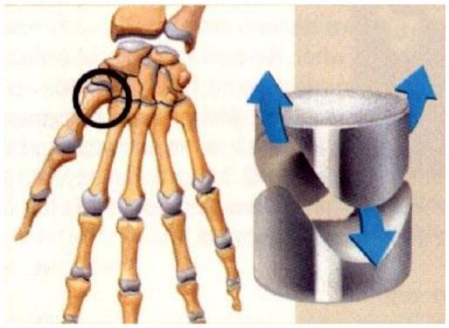
- Joint has 0-6 degrees of freedom (DoF): three coordinates and 3 rotation angles.

JOINTS AND DOF

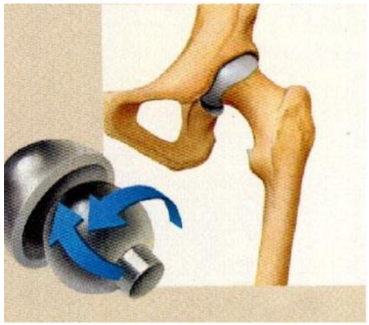
- Rotational joints
- Translational joints
- Compound (mix the previous two motions)



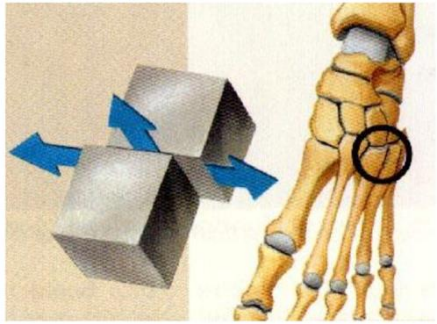
Hinge
Rotation along one axis
1DOF



Saddle
back and forth & up and down motion
2DOF



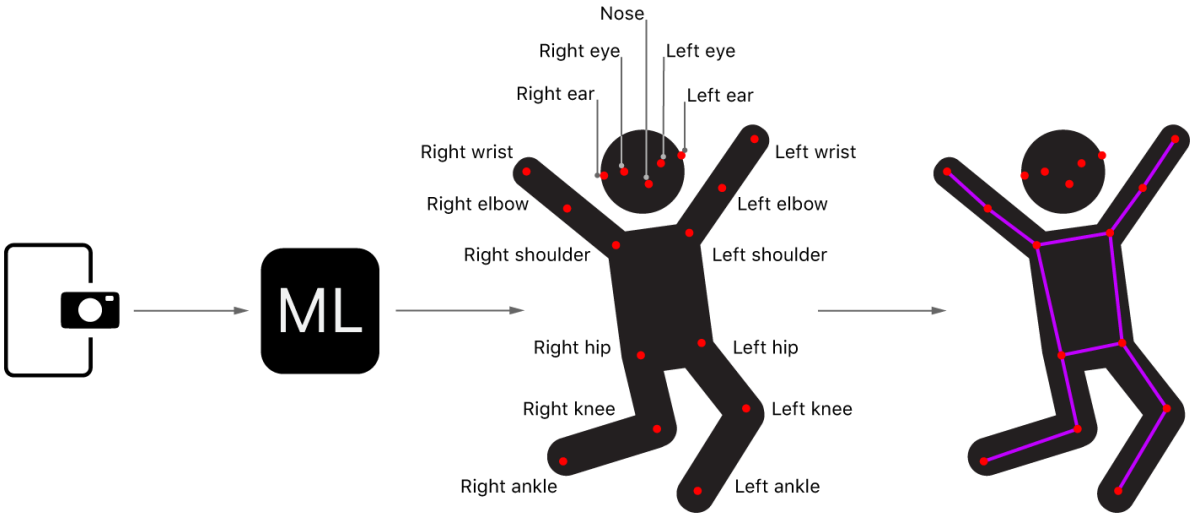
Ball-and-socket
Rotation along all 3 axes
3DOF



Sliding
Translation on a plane
2DOF

JOINTS FROM ACQUIRED SIGNALS

- Usually time series of 2D coordinates (n=2)
- One time serie for each joint; traditionally, 15 markers are used for a 30-dimensional time serie



- Acquired via different means
  i. Giroscopic sensors, accelerometers, … (IMU) -> specil equipment for diagnostics or information leackage from devices
  ii. RGB cameras
  iii. Depth cameras



https://www.nbc.com/saturday-night-live/video/motion-capture/n13489